Bayesian Batting Average Estimation: A Beta-Binomial Approach to Small-Sample Performance

Abstract:

Batting average is commonly used to evaluate hitting performance in baseball, but it does not account for sample size. This paper applies a beta-binomial Bayesian model to estimate a player's true batting ability from observed hit totals. Using a *Beta(25,75)* prior and simulated batting lines with the same naïve average but different numbers of at-bats, the model demonstrates how small samples are shrunk toward the prior while larger samples move closer to the observed average. The results show that Bayesian estimation proves a more cautious and informative approach to early-season or low-sample batting statistics by combining point estimates with credible intervals.

Introduction:

The normal metric for calculating batting average in baseball, in this paper referred to as a *naïve average*, is found using the formula:

$$AVG = \frac{hits}{at\ bats}$$

That is: the ratio of the number of times a player made contact with the ball to the number of chances they had. However, this particular statistic ignores sample size. For example, a player going 6/10 and 600/1000 both have a naïve average of *0.600*, but they are not equally convincing. A player going 6/10 is a good weekend of baseball, one going for 600/1000 is the best hitting season of all time.

Oftentimes, earlier in the season or in leagues where fewer games are played (such as the U.K. baseball leagues) naïve average is used as justification for a player's hitting ability when the sample size is far too low to make any serious judgment, i.e. 'he had a good weekend'. So, this paper considers a better statistic for a player's true hitting ability in these cases, attempting to answer the question: "*How does Bayesian estimation handle small-sampling batting averages compared to naïve batting average*".

Mathematical Background:

This experiment aimed to produce a better estimate of a player's true batting ability, that being their true probability of getting a hit, which is labelled as p. As mentioned, the traditional naïve average, $\hat{p}$, is calculated via:

$$\hat{p} = \frac{h}{n}$$

p = true probability of a hit, $\hat{p}$ = naïve batting average, h = observed hits, n = observed at bats

So, Naïve batting average is an attempt to estimate p, it is not the same thing as p. Clearly, a better calculation is necessary as $\hat{p}$, makes no consideration of sample size. Again:

$$\frac{6}{10} = 0.600$$

$$\frac{600}{1000} = 0.600$$

These have the same observed average, but hugely different amounts of evidence. So, it is untrue to argue that both of these hitters have the same ability, a method considering sample size and uncertainty must be used.

At-Bats were assumed to be independent of each other and that the same probability of getting a hit was present each plate appearance. Therefore, given the binary nature of an at-bat (that being a hitter either *gets a hit* or is *gets out*), at-bats were modelled as a Bernoulli Trial:

$$X \sim Bernoulli(p)$$

Where:

$$X = 1 \; means \; a \; hit$$

$$X = 0 \; means \; an \; out$$

Now, given the assumptions provided, if a player has true hit probability p, then the number of hits over n at-bats can be modelled using a Binomial distribution:

$$H \mid p \sim Binomial(n, p)$$

H = number of 'new' hits achieved, n = number of 'new' at bats

Bayesian statistics fits our new constraints as it represents uncertainty over p using a distribution. It allows us to use our prior data, the prior, to calculate the updated p given new data. Bayes' Theorem:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

$$\therefore P(p \mid data) \propto P(data \mid p)P(p)$$

Hence, the posterior combines what was believed before seeing the new data with how well different values of p explain the observed data. Making it a much stronger estimation for a true prediction of a player's hitting ability.

Since p is a probability which lies from $0 \leq p \leq 1$, a Beta distribution is used as a natural prior for the unknown probability:

$$p \sim Beta(\alpha, \beta)$$

α = *observed previous hits,* β = *observed prior outs*

That being, the model used for previously held information on a player's hitting, upon which we add the new data. The mean for a Beta distribution is calculated using:

$$E[p] = \frac{\alpha}{\alpha + \beta}$$

(α *+* β*) = hits + outs = total at-bats*

This experiment used a simple prior of

$$Beta(25,75)$$

That being one centred around 0.250, a common MLB average, which acts as 100 prior pseudo-at-bats.

So, when trying to update our estimate for *p*, the update rule adds newly observed hits to α and newly observed outs to β.

$$p \mid h, n \sim Beta(\alpha + h, \beta + n - h)$$

h = newly observed hits, (n-h) = newly observed outs

Ultimately, the value we will actually use will be our posterior mean, a more useful single-number estimate of the Bayesian average. It lies between the prior expectation and the observed average. Small samples stay close to the prior, large samples move closer to the observed average. This pulling toward the prior is called shrinkage.

$$E[p \mid h, n] = \frac{\alpha + h}{\alpha + \beta + n}$$

Moreover, we can further the accuracy of our estimate by measuring uncertainty. The posterior mean gives a single number which is useful for initial observation; however, given the posterior is a distribution it also gives a range of plausible values. This experiment used a 95% credible interval, that being the interval in which the central 95% of posterior probability is contained. Overall, larger samples should produce narrower credible intervals because the model has more evidence i.e. is more confident.

Method:

This experiment utilised Python to for calculations, accessible at:

src/bayesian_batting.py

The key functions written were those to calculate naïve average, posterior parameters, the posterior mean and the credible interval. Throughout this experiment the prior mentioned above was used for all examples:

$$Beta(25,75)$$

This was chosen as it is centred at 0.250 and behaves like 100 prior at-bats, both reasonable priors for all cases tested.

There was also a function *batting_summary* used which calculated all outputs required for one input/player. Using this function, the same batting average of 0.600 was tested across increasing sample sizes, those being:

Index	Hits	At-Bats
0	3	5
1	6	10
2	30	50
3	60	100
4	300	500

Figure 1 – A table illustrating the cases tested

Results:

From the cases tested, the following table was produced, displaying the results of each of the statistics measured in each case:

Experiment Number	Hits	At Bats	Naive Average	Posterior Mean	Posterior std	Credible Interval Lower	Credible Interval Upper	Credible Interval Width
0	3	5	0.6	0.267	0.043	0.187	0.355	0.168
1	6	10	0.6	0.282	0.043	0.202	0.369	0.167
2	30	50	0.6	0.367	0.039	0.292	0.445	0.153
3	60	100	0.6	0.425	0.035	0.357	0.494	0.137
4	300	500	0.6	0.542	0.020	0.502	0.581	0.079

Figure 2 – Table displaying all calculations for each of the cases tested

Whilst all of the data is displayed here, graphs were plotted using this information to make the analysis more intuitive.

Firstly, a graph was plotted to display change over number of at bats for both Naïve batting average and the new Bayesian estimate:

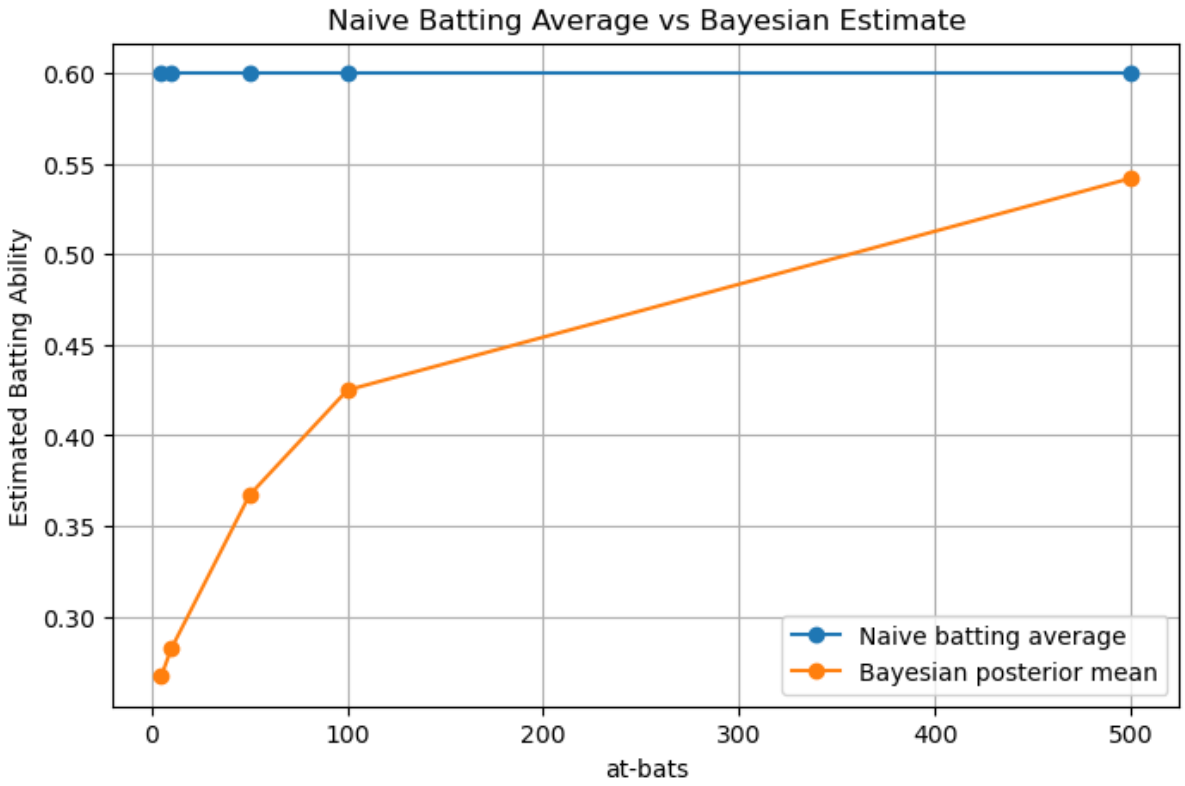


Figure 3 – A graph showing naïve batting average vs the Bayesian estimate

From which, it can be observed that, whilst the naïve batting average remains at 0.600 regardless of the number of at-bats taken, the Bayesian estimate starts much lower and moves toward 0.600 as the sample size increases. That is, as the confidence in p=0.600 increases, the Bayesian estimate moves towards such a value.

Next, a graph was plotted to display the change in posterior uncertainty as the sample size increases:

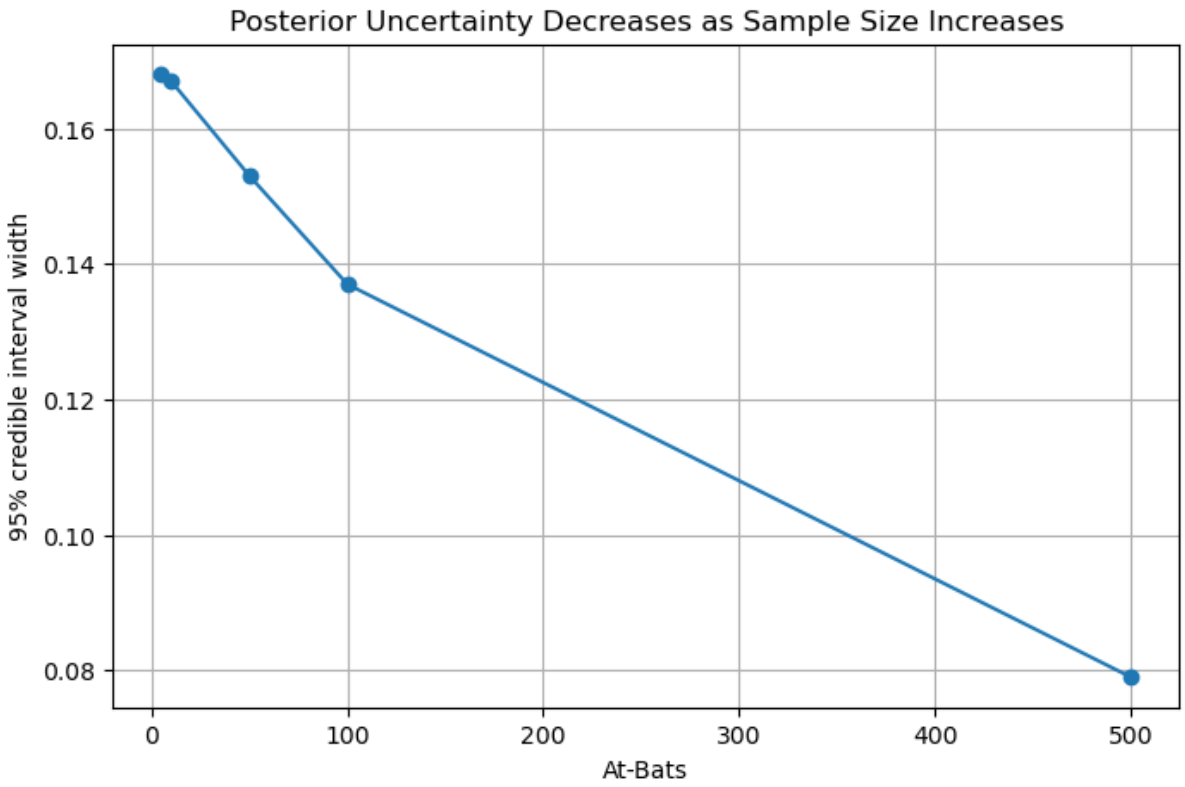


Figure 4 – A graph showing that posterior uncertainty decreases as sample size increases

It follows that, as sample size increases, the credible interval becomes narrower, meaning the model becomes more certain about the player's true batting ability.

Finally, a graph was plotted to visualise how the prior changes vs the posterior for one example. In other words, how does adding new data affect the Bayesian batting average we're modelling:

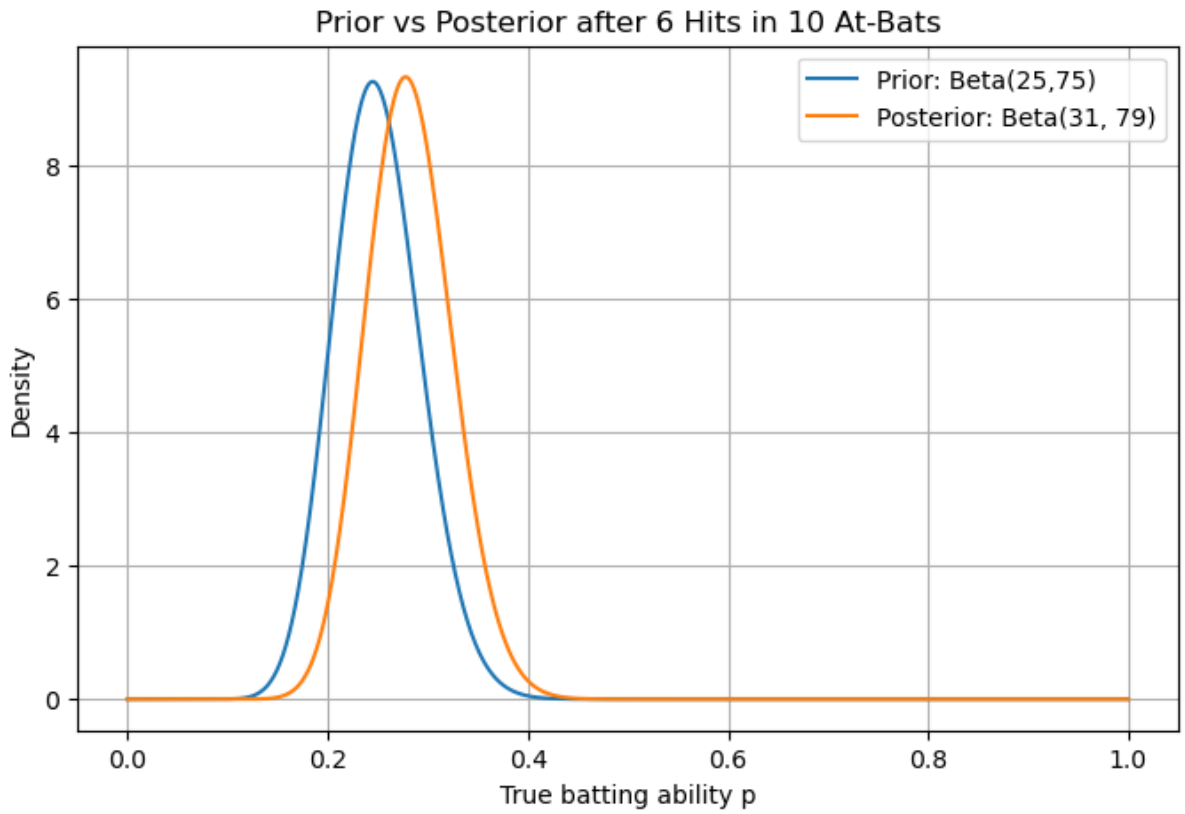


Figure 5 – A graph showing Prior vs Posterior average as new data was added.

From which we can see, after a small sample size increase, the posterior average shifts to the right. The new data added was that of 6 hits in 10 at-bats, a relatively small sample size; so, the posterior average increases slightly; however, it does not move all the way to 0.600 as the sample size is too small.

Conclusion:

This paper has highlighted the inadequacy of the Naïve batting average statistic as it ignores sample size. The demonstrated, beta-binomial, model shrinks small samples towards a prior and larger samples move closer to the observed average with narrower credible intervals. This makes the proposed method better suited for early-season statistics or in low-sample baseball contexts.

However, this experiment did have limitations. Each at-bat was modelled as independent which is not entirely true in game-scenarios, a player may for example be in a ‘slump’, struggle against a specific pitcher for multiple at bats, or face various other issues pertaining to the mental side of playing a sport. Moreover, this experiment only used one fixed prior which narrowed the set of results available to be analysed, specifically those pertaining to different statistical scenarios modelling different parts of the season. This experiment did not account for pitcher quality, league quality, ballpark factors, injuries or tactical hitting role. Moreover, only naïve batting average was considered, not more credible stats such as OBP/SLG/wOBA.

Future work should look to test different priors, use real season or player data, extend to the aforementioned, better, statistics and compare these estimates with future performance.